Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Dual-Stream Asynchronous Analysis for Multimodal Deepfake Detection

Anish Thakur, Aastik Sharma, Dhruv Garg

2210991273 | 2210991132 | 2210990273

Supervised By

Dr. Isha Kansal, Associate Professor, Chitkara University

Table of Contents

Introduction

CHITKARA UNIVERSITY

- Deepfakes are AI-generated synthetic media created using deep learning techniques such as GANs and autoencoders.
- They pose a dual-domain threat — both video and audio streams can be independently manipulated to spread misinformation.
- Existing detection methods are largely unimodal, focusing only on visual or audio forensics in isolation.
- A single-stream approach fails when only one modality is forged — e.g., real video paired with fake audio.
- This paper proposes a parallel-path multimodal framework that simultaneously analyzes both video and audio streams for robust detection.

Problem Statement

CHITKARA UNIVERSITY

Key Challenges in Deepfake Detection:

- Rapid advancement of generative models (StyleGAN, Diffusion Models) produces near-perfect forgeries.
- Most existing systems are unimodal — analyzing only video frames OR only audio waveforms.
- Unimodal systems are easily defeated when only one stream is manipulated.

Research Gap:

- Limited research on multimodal, fusion-based detection that evaluates both audio and visual streams simultaneously.
- No robust stacked ensemble approach combining spatial, spatio-temporal, and audio models.

Objective:

- Design a highly accurate (99%+) multimodal detection system using ensemble learning and fusion logic.

Proposed Multimodal Detection Framework

CHITKARA UNIVERSITY

System Pipeline:

- Input: A video file containing both audio and visual content.
- Demultiplexing: FFmpeg separates the file into Stream 1 (Video Frames) and Stream 2 (Audio .wav).
- Parallel Processing: Both streams are processed simultaneously and independently.

Dual-Stream Architecture:

- Stream 1 → Video Detection Module → Probability Score: P_video
- Stream 2 → Audio Detection Module → Probability Score: P_audio

Fusion Logic:

- $P\_final = (0.6 \times P\_video) + (0.4 \times P\_audio)$
- Weighted average favors video module (higher tested accuracy); robust even if one modality is ambiguous.

Video Detection Module (Stacked Ensemble)

Preprocessing:

- Video frames extracted at constant rate; MTCNN face detector crops and resizes facial regions to 224×224.

Base Models (Level 0):

- EfficientNet-B0 (Spatial): High accuracy, identifies fine-grained texture artifacts and pixel inconsistencies.
- ResNet50 (Spatial): Robust deep residual network; validated accuracy 98.39%; powerful feature extractor.
- CNN+LSTM (Spatio-Temporal): TimeDistributed(MobileNetV2) + LSTM; detects unnatural blinking, rigid lip movement.

Meta-Learner (Level 1 Stacking):

- A shallow neural network (Dense 16 → ReLU → Dense 1 → Sigmoid) trained on base model predictions.
- Learns optimal non-linear combination of base models — when to trust each model.
- Final ensemble accuracy on Celeb-DF test set: 99.1%

Audio Detection Module (ResNet50)

Preprocessing Pipeline:

- Raw audio waveform converted to a Mel Spectrogram — a 2D time-frequency representation.
- Spectrogram normalized to 224×224 pixels to match ResNet50 input requirements.
- Single-channel spectrogram replicated to 3-channel (RGB) format for CNN compatibility.

Model Architecture:

- Base: ResNet50 pre-trained on ImageNet — powerful feature extractor for spectrogram images.
- Custom head: AveragePooling2D → Flatten → Dense(128, ReLU) → Dropout(0.5) → Dense(1, Sigmoid).

Training Strategy:

- Phase 1 (10 epochs): Train classifier head only; ResNet base frozen.
- Phase 2 (5 epochs): Fine-tune full model (base unfrozen) for domain adaptation.
- Trained on LJSpeech (real) + WaveFake (fake) datasets. Validation Accuracy: 98.39%.

Experiments & Results

Datasets:

- Video: FaceForensics++ (FF++) for training; Celeb-DF for robustness testing.
- Audio: LJSpeech (real corpus) + WaveFake (fake corpus).

Implementation:

- Optimizer: Adam (lr=0.0001); Loss: Binary Cross-Entropy; Hardware: GPU-accelerated.

Video Module Performance (Celeb-DF Test Set):

- ResNet50 (baseline): 98.1% accuracy | EfficientNet-B0: ~97% | CNN+LSTM: ~96%
- Stacked Ensemble Meta-Learner: 99.1% accuracy (+1% absolute improvement over best base model)

Audio Module Performance:

- ResNet50 on spectrograms: Validation Accuracy 98.39%; smooth convergence over 15 epochs.

Final Multimodal System:

- Combined Audio-Visual Test Accuracy: 99.4% | Confusion Matrix (10,000 samples): 4970 TN, 4970 TP, 30 FP, 30 FN

Conclusion & Future Work

CHITKARA UNIVERSITY

Conclusion:

- Proposed a novel dual-stream multimodal deepfake detection framework combining ensemble video analysis with spectro-temporal audio detection.
- The stacked ensemble video module (EfficientNet + ResNet50 + CNN-LSTM) achieves 99.1% test accuracy.
- The audio ResNet50 module achieves 98.39% validation accuracy on LJSpeech/WaveFake datasets.
- The fused multimodal system achieves state-of-the-art 99.4% accuracy, successfully detecting cross-modal forgeries.

Future Work:

- Extend the framework to handle real-time streaming video analysis.
- Incorporate transformer-based architectures (ViT, Audio Spectrogram Transformers) for further accuracy gains.
- Develop adversarially robust models to counter adaptive forgery attacks.
- Explore generalization across diverse deepfake generation techniques and datasets.

References

- [1] S. Das et al., “Enhanced deepfake detection using CNN and EfficientNet ensemble models,” ICACSE 2025.
- [2] A. Jaiswal et al., “A novel framework for deepfake image detection using deep learning,” L&T 2025.
- [3] A. Mahar et al., “Detection of quality deepfake images using customised CNNs,” ICACSE 2025.
- [4] V. Shinde et al., “Audio deepfake detection using CNN and Bi-LSTM,” ICRMLAI 2024.
- [5] K. He et al., “Deep Residual Learning for Image Recognition,” CVPR 2016, pp. 770-778.
- [6] M. Tan & Q. Le, “EfficientNet: Rethinking Model Scaling for CNNs,” ICML 2019.
- [7] A. Rossler et al., “FaceForensics++: Learning to Detect Forged Facial Images,” ICCV 2019.
- [8] S. Hochreiter & J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, 1997.
- [9] Y. Li et al., “Celeb-DF: A Large-Scale Challenging Dataset for Deepfake Forensics,” CVPR 2020.
- [10] K. Ito & L. Johnson, “The LJ Speech Dataset,” 2017.
- [11] J. Frank & L. Schonherr, “WaveFake: A Data Set to Detect Deepfake Audio,” arXiv 2021.
- [12] D. Kingma & J. Ba, “Adam: A Method for Stochastic Optimization,” arXiv 2014.

Thank You